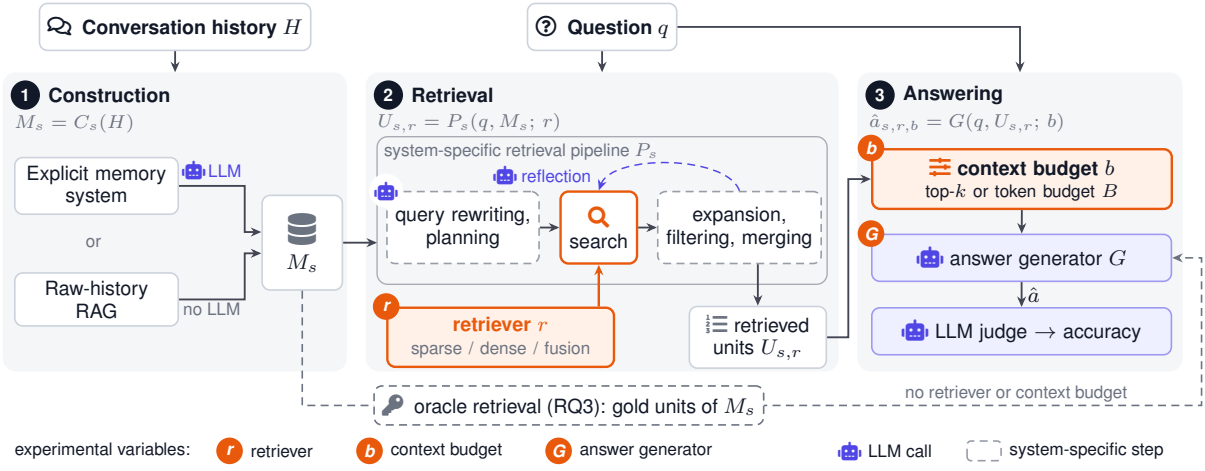

Conversation history H
Question q
1 Construction
$M_s = C_s(H)$
Explicit memory system
LLM
or
Raw-history RAG
no LLM
M_s
2 Retrieval
$U_{s,r} = P_s(q, M_s;\ r)$
system-specific retrieval pipeline P_s
reflection
query rewriting, planning
search
expansion, filtering, merging
r
retriever r
sparse / dense / fusion
retrieved units $U_{s,r}$
3 Answering
$\hat{a}_{s,r,b} = G(q, U_{s,r};\ b)$
b
context budget b
top-k or token budget B
G
answer generator G
$\hat{a}$
LLM judge → accuracy
oracle retrieval (RQ3): gold units of M_s
no retriever or context budget
experimental variables:
r retriever
b context budget
G answer generator
LLM call
system-specific step